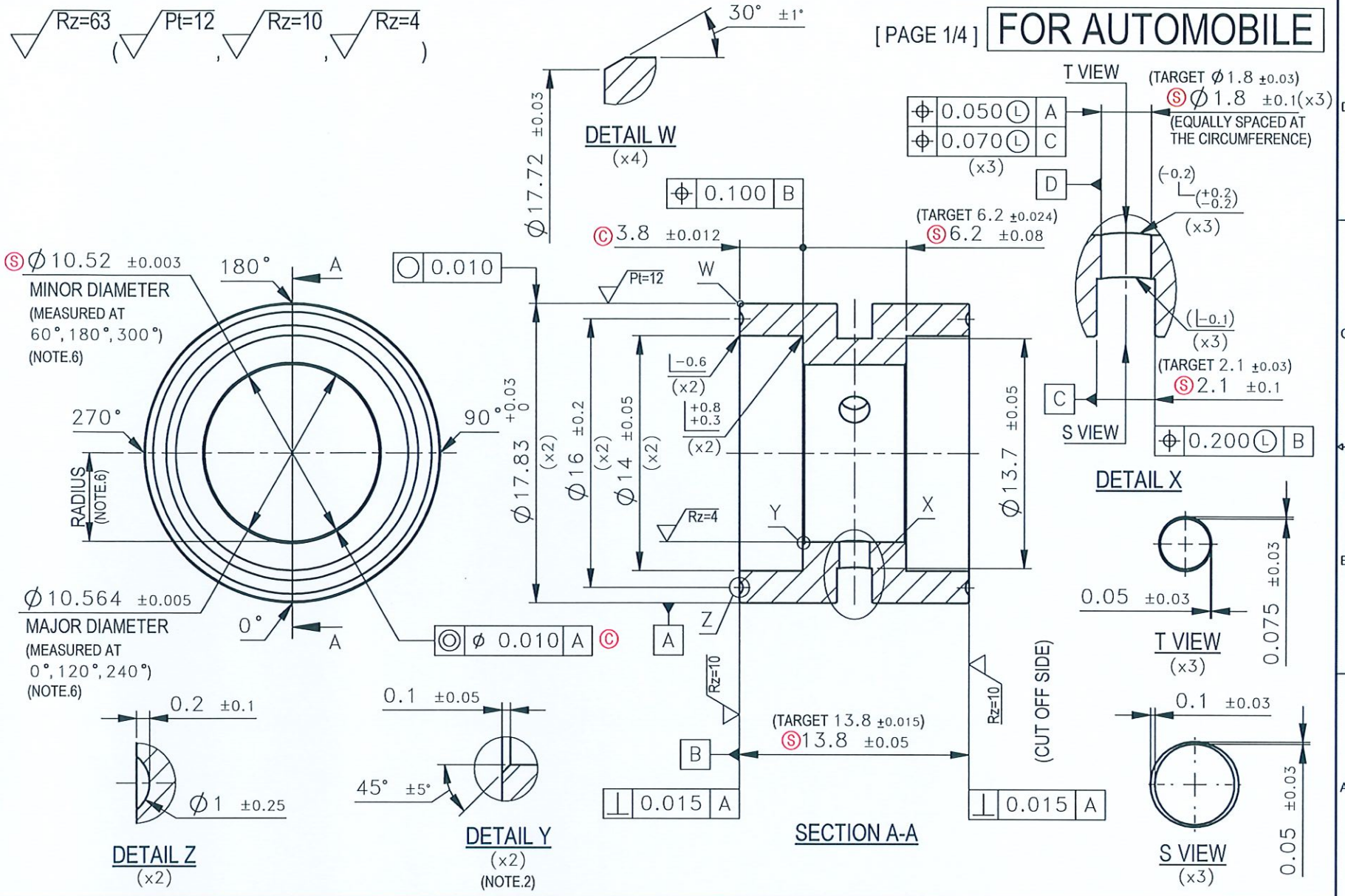
Rz=63 (Pt=12, Rz=10, Rz=4)

[PAGE 1/4] **FOR AUTOMOBILE**



PROCESS No.	
T7099AM1A	NR REVISION
ITEM NAME	SLEEVE
PART No.	OT-7099A Rev.A
PROCESS	① TURNING
CONTAINER	PIN1+PUNCH

No.	PROCESS NAME.
①	TURNING
②	CGM
◯	SHIP

ISSUED
5 Sep'22
Precision Mechanical Assembly Div.

Ⓢ CUSTOMER CRITICAL POINT
Ⓒ CRITICAL POINT

LASTEST ECN.				MACHINE
REV.	ECN No.	DATE	NAME	B0326
				-
				-
				-
				-

MATERIAL TYPE	MAT'L SIZE
SD4	Ø18.3
HARDNESS	
-	
SURFACE COATING	
-	

GENERAL TOLERANCE	
LESS THAN 0.5	±0.05
0.5 - 6	±0.1
OVER 6 - 30	±0.2
OVER 30 - 120	±0.3
OVER 120	±0.5
ANGLE	±1°

UNLESS SPECIFIED
CONVEX = C0.2
RECESS = R0.2 MAX

3rd. ANGLE

UNIT : mm.

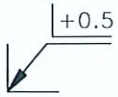
NMB
MinebeaMitsumi Inc.
Precision Mechanical Assembly Division

DETAILED	CHECKED	APPROVED
WORACHET		

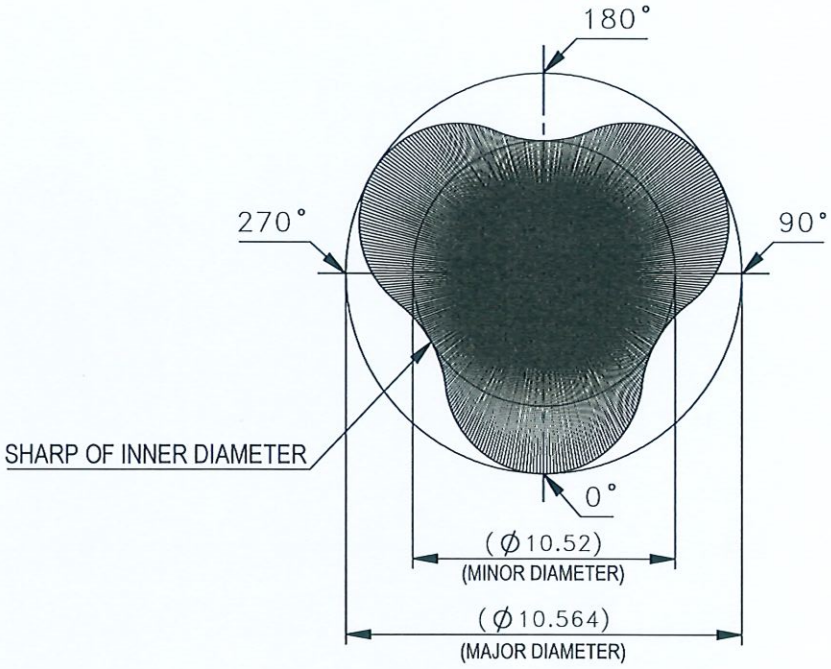
DATE: 1 SEP '2022

[PAGE 2/4] FOR AUTOMOBILE

NOTE : 1.PART TO BE FREE OF BURRS, CHIPS AND CONTAMINANTS.
2.THE CHAMFER OF INNER DIAMETER IS ALLOWED A CIRCLE TYPE.
IN THIS CASE, THE MEASUREMENT POINT IS ∅10.52.
3.DIMENSIONS WITHOUT TOLERANCES ACCORDING TO ISO 2768-mK.
4.WITHOUT INSTRUCTION A CORNER.



5.PART A/RUN SHALL DIP IN RUSTY PREVENTION SOLVENT.
6.PROFILE OF A SURFACE AREA IS MEASURE IN THE SECTION ① TO ⑥.
(FIGTURE.1 AND TABLE.1)



SHARP OF INNER DIAMETER
(FIGTURE.1)

PROCESS No.	NR REVISION
T7099AM1A	
ITEM NAME: SLEEVE	
PART No.: OT-7099A Rev.A	
PROCESS: ① TURNING	
CONTAINER: PIN1+PUNCH	

No.	PROCESS NAME.
①	TURNING
②	CGM
◯	SHIP

ISSUED
5 Sep'22
Precision Mechanical Assembly Div.

Ⓢ CUSTOMER CRITICAL POINT
Ⓒ CRITICAL POINT

LASTEST ECN.				MACHINE	MATERIAL TYPE	MAT'L SIZE
REV.	ECN No.	DATE	NAME	B0326	SD4	∅18.3
				-	HARDNESS	
				-	-	
				-	SURFACE COATING	
				-	-	

GENERAL TOLERANCE	
LESS THAN 0.5	±0.05
0.5 - 6	±0.1
OVER 6 - 30	±0.2
OVER 30 - 120	±0.3
OVER 120	±0.5
ANGLE	±1°

UNLESS SPECIFIED
CONVEX = C0.2
RECESS = R0.2 MAX

3rd. ANGLE
UNIT : mm.

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Precision Mechanical Assembly Division

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WORACHET		

DATE: 1 SEP '2022

TABLE.1 : PROFILE OF A SURFACE AREA IS MEASURE IN THE SECTION ① TO ⑥.

[PAGE 3/4]

FOR AUTOMOBILE

ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE
0	5.2820		45	5.2632		90	5.2710		135	5.2788	
1	5.2820		46	5.2628		91	5.2716		136	5.2784	
2	5.2819		47	5.2625		92	5.2721		137	5.2779	
3	5.2819		48	5.2621	① (NOTE.6)	93	5.2727		138	5.2775	
4	5.2818		49	5.2618	⌓ 1.194 A B DⓂ	94	5.2733		139	5.2770	
5	5.2816		50	5.2615	⌓ 0.010 A B	95	5.2738		140	5.2765	
6	5.2815		51	5.2612		96	5.2744		141	5.2760	
7	5.2813		52	5.2610		97	5.2749		142	5.2755	
8	5.2810		53	5.2607		98	5.2755		143	5.2749	
9	5.2808		54	5.2605		99	5.2760		144	5.2744	
10	5.2805		55	5.2604		100	5.2765		145	5.2738	
11	5.2802		56	5.2602		101	5.2770		146	5.2733	
12	5.2799		57	5.2601		102	5.2775		147	5.2727	
13	5.2795		58	5.2601		103	5.2779		148	5.2721	
14	5.2792		59	5.2600	② (NOTE.6)	104	5.2784		149	5.2716	③ (NOTE.6)
15	5.2788		60	5.2600	⌓ 1.194 A B DⓂ	105	5.2788		150	5.2710	⌓ 1.194 A B DⓂ
16	5.2784		61	5.2600	⌓ 0.010 A B	106	5.2792		151	5.2704	⌓ 0.010 A B
17	5.2779		62	5.2601		107	5.2795		152	5.2699	
18	5.2775		63	5.2601		108	5.2799		153	5.2693	
19	5.2770	① (NOTE.6)	64	5.2602		109	5.2802		154	5.2687	
20	5.2765	⌓ 1.194 A B DⓂ	65	5.2604		110	5.2805	③ (NOTE.6)	155	5.2682	
21	5.2760	⌓ 0.010 A B	66	5.2605		111	5.2808	⌓ 1.194 A B DⓂ	156	5.2676	
22	5.2755		67	5.2607		112	5.2810	⌓ 0.010 A B	157	5.2671	
23	5.2749		68	5.2610		113	5.2813		158	5.2665	
24	5.2744		69	5.2612		114	5.2815		159	5.2660	
25	5.2738		70	5.2615		115	5.2816		160	5.2655	
26	5.2733		71	5.2618		116	5.2818		161	5.2650	
27	5.2727		72	5.2621		117	5.2819		162	5.2645	
28	5.2721		73	5.2625		118	5.2819		163	5.2641	
29	5.2716		74	5.2628		119	5.2820		164	5.2636	
30	5.2710		75	5.2632	③ (NOTE.6)	120	5.2820		165	5.2632	
31	5.2704		76	5.2636	⌓ 1.194 A B DⓂ	121	5.2820		166	5.2628	
32	5.2699		77	5.2641	⌓ 0.010 A B	122	5.2819		167	5.2625	
33	5.2693		78	5.2645		123	5.2819		168	5.2621	
34	5.2687		79	5.2650		124	5.2818		169	5.2618	
35	5.2682		80	5.2655		125	5.2816		170	5.2615	
36	5.2676		81	5.2660		126	5.2815		171	5.2612	
37	5.2671		82	5.2665		127	5.2813		172	5.2610	
38	5.2665		83	5.2671		128	5.2810		173	5.2607	
39	5.2660		84	5.2676		129	5.2808		174	5.2605	
40	5.2655		85	5.2682		130	5.2805		175	5.2604	
41	5.2650		86	5.2687		131	5.2802		176	5.2602	④ (NOTE.6)
42	5.2645		87	5.2693		132	5.2799		177	5.2601	⌓ 1.194 A B DⓂ
43	5.2641		88	5.2699		133	5.2795		178	5.2601	⌓ 0.010 A B
44	5.2636		89	5.2704		134	5.2792		179	5.2600	

PROCESS No.
T7099AM1A
NR REVISION

ITEM NAME
SLEEVE

PART No.
OT-7099A Rev.A

PROCESS
① TURNING

CONTAINER
PIN1+PUNCH

No.	PROCESS NAME.
①	TURNING
②	CGM
◯	SHIP
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	

ISSUED
5 Sep'22
Precision Mechanical Assembly Div.

Ⓢ CUSTOMER CRITICAL POINT
Ⓒ CRITICAL POINT

LASTEST ECN.				MACHINE
REV.	ECN No.	DATE	NAME	B0326
				-
				-
				-
				-

MATERIAL TYPE: SD4

MAT'L SIZE: ∅18.3

HARDNESS: -

SURFACE COATING: -

GENERAL TOLERANCE	
LESS THAN 0.5	±0.05
0.5 - 6	±0.1
OVER 6 - 30	±0.2
OVER 30 - 120	±0.3
OVER 120	±0.5
ANGLE	±1°

UNLESS SPECIFIED
CONVEX = C0.2
RECESS = R0.2 MAX

3rd. ANGLE

UNIT : mm.

NMB
MinebeaMitsumi Inc.
Precision Mechanical Assembly Division

DETAILED	CHECKED	APPROVED
WORACHET		

DATE: 1 SEP '2022

TABLE.1 : PROFILE OF A SURFACE AREA IS MEASURED IN THE SECTION ① TO ⑥.

[PAGE 4/4] **FOR AUTOMOBILE**

ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE	ANGLE	RADIUS	PROFILE TOLERANCE
180	5.2600	④(NOTE.6)	225	5.2788		270	5.2710		315	5.2632	
181	5.2600	⌭ 1.194 A B Ⓜ	226	5.2792		271	5.2704		316	5.2636	
182	5.2601	⌭ 0.010 A B	227	5.2795		272	5.2699		317	5.2641	
183	5.2601		228	5.2799		273	5.2693		318	5.2645	
184	5.2602		229	5.2802		274	5.2687		319	5.2650	
185	5.2604		230	5.2805		275	5.2682		320	5.2655	
186	5.2605		231	5.2808		276	5.2676		321	5.2660	
187	5.2607		232	5.2810		277	5.2671		322	5.2665	
188	5.2610		233	5.2813		278	5.2665		323	5.2671	
189	5.2612		234	5.2815		279	5.2660	⑤(NOTE.6)	324	5.2676	
190	5.2615		235	5.2816		280	5.2655	⌭ 1.194 A B Ⓜ	325	5.2682	
191	5.2618		236	5.2818		281	5.2650	⌭ 0.010 A B	326	5.2687	
192	5.2621		237	5.2819		282	5.2645		327	5.2693	
193	5.2625		238	5.2819		283	5.2641		328	5.2699	
194	5.2628		239	5.2820		284	5.2636		329	5.2704	
195	5.2632		240	5.2820		285	5.2632		330	5.2710	
196	5.2636		241	5.2820		286	5.2628		331	5.2716	
197	5.2641		242	5.2819		287	5.2625		332	5.2721	
198	5.2645		243	5.2819		288	5.2621		333	5.2727	
199	5.2650		244	5.2818	⑤(NOTE.6)	289	5.2618		334	5.2733	①(NOTE.6)
200	5.2655		245	5.2816	⌭ 1.194 A B Ⓜ	290	5.2615		335	5.2738	⌭ 1.194 A B Ⓜ
201	5.2660		246	5.2815	⌭ 0.010 A B	291	5.2612		336	5.2744	⌭ 0.010 A B
202	5.2665		247	5.2813		292	5.2610		337	5.2749	
203	5.2671	⑤(NOTE.6)	248	5.2810		293	5.2607		338	5.2755	
204	5.2676	⌭ 1.194 A B Ⓜ	249	5.2808		294	5.2605		339	5.2760	
205	5.2682	⌭ 0.010 A B	250	5.2805		295	5.0604		340	5.2765	
206	5.2687		251	5.2802		296	5.2602		341	5.2770	
207	5.2693		252	5.2799		297	5.2601		342	5.2775	
208	5.2699		253	5.2795		298	5.2601	⑥(NOTE.6)	343	5.2779	
209	5.2704		254	5.2792		299	5.2600	⌭ 1.194 A B Ⓜ	344	5.2784	
210	5.2710		255	5.2788		300	5.2600	⌭ 0.010 A B	345	5.2788	
211	5.2716		256	5.2784		301	5.2600		346	5.2792	
212	5.2721		257	5.2779		302	5.2601		347	5.2795	
213	5.2727		258	5.2775		303	5.2601		348	5.2799	
214	5.2733		259	5.2770		304	5.2602		349	5.2802	
215	5.2738		260	5.2765		305	5.2604		350	5.2805	
216	5.2744		261	5.2760		306	5.2605		351	5.2808	
217	5.2749		262	5.2755		307	5.2607		352	5.2810	
218	5.2755		263	5.2749		308	5.2610	①(NOTE.6)	353	5.2813	
219	5.2760		264	5.2744		309	5.2612	⌭ 1.194 A B Ⓜ	354	5.2815	
220	5.2765		265	5.2738		310	5.2615	⌭ 0.010 A B	355	5.2816	
221	5.2770		266	5.2733		311	5.2618		356	5.2818	
222	5.2775		267	5.2727		312	5.2621		357	5.2819	
223	5.2779		268	5.2721		313	5.2625		358	5.2819	
224	5.2784		269	5.2716		314	5.2628		359	5.2820	

PROCESS No. T7099AM1A (NR REVISION)

ITEM NAME SLEEVE

PART No. OT-7099A Rev.A

PROCESS ① TURNING

CONTAINER PIN1+PUNCH

No.	PROCESS NAME.
①	TURNING
②	CGM
◯	SHIP
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	
◯	

Ⓢ CUSTOMER CRITICAL POINT

Ⓒ CRITICAL POINT

LASTEST ECN.				MACHINE
REV.	ECN No.	DATE	NAME	B0326
				-
				-
				-
				-

MATERIAL TYPE: SD4

MAT'L SIZE: ∅18.3

HARDNESS: -

SURFACE COATING: -

GENERAL TOLERANCE	
LESS THAN 0.5	±0.05
0.5 - 6	±0.1
OVER 6 - 30	±0.2
OVER 30 - 120	±0.3
OVER 120	±0.5
ANGLE	±1°

UNLESS SPECIFIED
CONVEX = C0.2
RECESS = R0.2 MAX

3rd. ANGLE

UNIT : mm.

NMB MinebeaMitsumi Inc.

Precision Mechanical Assembly Division

DETAILED	CHECKED	APPROVED
WORACHET		

DATE: 1 SEP '2022